

Hollow-Core and Standard Single-Mode Fiber Hybrid Optical Networks: A Multi-Topology Investigation of Protection Switching

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Abstract We study cross-fiber protection switching in hybrid hollow-core and single-mode networks. At 50% deployment, chromatic dispersion (CD) steps reach 4,000–22,000 ps/nm with 1.6–3.1 dB penalties, causing 38–59% modulation downgrades. CO₂ absorption degrades 9–26 of 64 L-band channels. Higher HCF fractions generally reduce all impairments across six network topologies.

Introduction

Hollow-core fibers (HCFs) based on the antiresonant nodeless design [1] have matured from laboratory curiosities into deployed infrastructure. Record attenuations below 0.1 dB/km [2], near-zero Kerr nonlinearity, and ~30% lower latency make HCF attractive for backbone upgrades [3, 4]. Microsoft now operates over 1280 km of HCF in production Azure data-center interconnects [5].

Recent network-level planning studies confirm that selective HCF deployment delivers disproportionate gains. Ibrahim et al. [6] showed that upgrading just 13.7% of links to HCF reduces edge data-center requirements by 24–50%. Pedro et al. [7] demonstrated that allocating 10–20% of fiber spans to HCF in a pan-European topology increases feasible 800G lightpaths by 36–100%. These studies collectively establish that *hybrid* HCF–SMF networks—where working and protection paths may traverse different fiber types—are the realistic near-term deployment scenario.

In such hybrid networks, 1+1 protection switching under ITU-T G.808.1 may route working and protection paths over different fiber types. This cross-fiber switching introduces challenges absent in conventional all-SMF networks. The five times dispersion coefficient contrast creates CD steps of thousands to tens of thousands of ps/nm, requiring substantially larger equalizer adaptation ranges [8]. HCF exhibits intermodal interference (IMI) at approximately –55 dB/km [3], which is absent in SMF. Furthermore, HCF’s negligible nonlinearity permits per-channel launch powers up to 15 dBm (~33 dBm total for 64 channels) versus ~0 dBm for SMF, creating large GSNR asymmetries during switchover. Additionally, narrowband CO₂ absorption lines within the hollow core can degrade specific WDM channels [9–12], an impairment unique to HCF adding complexity to protection switching.

No prior study has quantified these cross-fiber effects at the network level to the best of our knowledge. We address this gap by evaluating protection switching across six reference topologies totalling 1,602 node pairs, using Monte Carlo simulation with random per-link fiber assignment to capture realistic mixed-path scenarios.

System Model

Fiber Parameters and Link Model

Table 1 summarizes the fiber parameters. Each topology uses link distances from the standard reference definitions (e.g., NSFNET links range from 350 to 2,800 km; Nobel-Germany links from 38 to 382 km). Each link is modelled as an optical multiplex section (OMS) subdivided into uniform amplifier spans of $L_{\text{span}} = 80$ km for SMF or 100 km for HCF, with inline EDFAs and 64 WDM channels per band at 75 GHz spacing with 64 GBaud symbol rate. Links shorter than one span length are modelled as a single span of the actual link distance.

Tab. 1: Fiber Parameters Used in the Study (C-band / L-band)

Parameter	SMF (C/L)	HCF (C/L)
α [dB/km]	0.20 / 0.21	0.13 / 0.14
D [ps/(nm·km)]	17.0 / 20.0	3.5 / 4.0
γ [W ⁻¹ km ⁻¹]	1.3 / 1.15	0.001
EDFA NF [dB]	5.5 / 6.5	5.5 / 6.5
IMI [dB/km]	—	–55 / –53
n_{eff}	1.468	1.0003

GSNR Model

The end-to-end generalized signal-to-noise ratio (GSNR) for each path includes ASE noise, Kerr nonlinear interference (NLI), and intermodal interference (IMI) [3]:

$$\text{GSNR} = \frac{P_{\text{ch}}}{P_{\text{ASE}} + P_{\text{NLI}} + P_{\text{IMI}}} \quad (1)$$

NLI is computed via the GN model [13]; for HCF, $\gamma \approx 0$ yields negligible NLI. IMI accumulates linearly with HCF length at –55 dB/km (C-band) and is a minor but non-negligible contributor over long HCF paths [3].

For each link, the per-channel launch power is set to the locally optimized (LOGO) value [3]:

$$P_{\text{opt}} = \left(\frac{P_{\text{ASE}}}{2\eta} \right)^{1/3} \quad (2)$$

where η is the per-span nonlinear efficiency from the GN model. Since HCF has $\eta \rightarrow 0$, its per-channel launch power is clamped at 15 dBm versus ~0 dBm for SMF, creating a ~15 dB per-channel power contrast.

For heterogeneous paths, the end-to-end GSNR is the harmonic sum of individual link GSNRs: $\text{GSNR}_{\text{path}}^{-1} = \sum_{\ell} \text{GSNR}_{\ell}^{-1}$, where each link receives its own LOGO launch power.

For L-band, GSNR is computed independently using L-band parameters from Table 1 at a reference wavelength of 1590 nm. CO₂ absorption in HCF produces narrowband notches [11, 14] with reported peak values in the range ~0.02–0.08 dB/km and maximum peak values of ~0.08–0.2 dB/km [10, 11, 15]. For channels whose wavelength falls within a notch, the corresponding excess attenuation is treated as an additional per-channel GSNR penalty and subtracted from the L-band baseline.

CD Step and FDE Tap Count

Upon switchover, the instantaneous CD step is:

$$\Delta\text{CD} = \left| \sum_{\ell \in W} D_{\ell} L_{\ell} - \sum_{\ell \in P} D_{\ell} L_{\ell} \right| \quad (3)$$

where W and P denote the sets of links on the working and protection paths, and each link ℓ may have either

D_{SMF} or D_{HCF} . The number of $T/2$ -spaced FDE taps required to compensate a given CD value is [8]:

$$N_{\text{taps}} = \left\lceil \frac{|\text{CD}| \cdot \Delta\lambda_{\text{sig}}}{T/2} \right\rceil + 1 \quad (4)$$

where $\Delta\lambda_{\text{sig}} = \lambda^2 R_s / c$ with R_s being the symbol rate. Applying Eq. (4) to ΔCD from Eq. (3) gives the additional FDE taps the equalizer must acquire upon switchover. At 64 GBaud with $2\times$ oversampling, every 1,000 ps/nm of CD step requires ~ 66 additional taps.

Modulation Format Selection

The achievable modulation format is determined by comparing end-to-end GSNR against standard thresholds with system margin: DP-64QAM (≥ 24 dB), DP-16QAM (≥ 18 dB), DP-8QAM (≥ 14 dB), and DP-QPSK (≥ 11 dB).

Network Topologies and Path Computation

We evaluate six standard reference topologies: NSFNET (14/21), Nobel-Germany (17/26), COST 239 (11/26), USNET (24/43), COST 266 (37/57), and CORONET (30/48), where the notation is (nodes/links). For each of the 1,602 node pairs with edge-disjoint paths, working and protection paths are computed using Suurballe's algorithm [16], which finds the minimum-cost edge-disjoint path pair. By construction, the working path is always the shorter of the two disjoint paths, and the protection path is typically longer due to the edge-disjoint constraint—in extreme cases (e.g., Seattle–Portland in USNET: 280 km working vs. 3,300 km protection), the protection path must traverse the entire backbone to avoid sharing any links with the working path.

Monte Carlo Simulation Methodology

Rather than assigning entire paths to a single fiber type (an extreme scenario unlikely in practice), we adopt a per-link Monte Carlo approach that captures realistic mixed-path deployments. At each HCF deployment fraction $f \in \{0, 0.1, \dots, 1.0\}$, each link in the topology is independently assigned as HCF with probability f (otherwise SMF). For each random realization, Suurballe's algorithm finds the working and protection paths through the mixed network, and the physical-layer metrics (CD step, GSNR penalty, FDE taps, modulation format) are computed link-by-link.

We use 200 independent trials per deployment fraction. A convergence analysis on NSFNET (21 links, 91 pairs) and CORONET (48 links, 378 pairs) showed that at 200 trials the standard error of the mean CD step is $< 1.5\%$ relative to the converged value, and the 95th-percentile error is $< 2\%$.

Results and Discussion

CD Step and GSNR Penalty Distributions (C-Band)

Figure 1 shows CDFs of CD step and GSNR penalty from the Monte Carlo mixed-path simulation at 25%, 50%, and 75% HCF deployment. At 50% HCF, mean CD steps range from 4,078 ps/nm (Nobel-Germany) to 21,947 ps/nm (NSFNET). The CD step magnitude is governed by path length: NSFNET's long inter-node distances (mean link 1,095 km) produce the largest CD steps, while Nobel-Germany's compact regional topology (mean link 186 km) yields the smallest. Increasing the HCF fraction from 25% to 75% progressively reduces the CD step, as more links transition from $D_{\text{SMF}} = 17$ to $D_{\text{HCF}} = 3.5$ ps/(nm km). The mean GSNR penalty (Fig. 1(b)) is positive (1.6–3.1 dB) across all topologies because Suurballe's algorithm assigns the shorter path as the working path, which on average achieves higher GSNR. However, 19–34% of individual node-pair samples exhibit negative penalties, indicating cases where

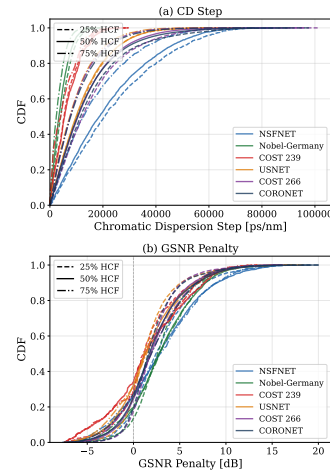


Fig. 1: C-band CDF of (a) chromatic dispersion step and (b) GSNR penalty across six reference topologies under Monte Carlo mixed-path deployment at 25%, 50%, and 75% HCF fraction (200 trials each). Dashed, solid, and dash-dot lines distinguish the three fractions.

Tab. 2: Monte Carlo Mixed-Path Results at 50% HCF (200 Trials)

Topology	Samples	ΔCD [ps/nm]	$\bar{N}_{\text{taps}}$	GSNR pen. [dB]	Mod. ↓ [%]
NSFNET	18,200	21,947	1,442	+3.1	59
Nobel-Ger.	27,200	4,078	269	+3.0	39
COST 239	11,000	6,678	440	+1.6	38
USNET	55,200	12,176	801	+1.7	42
COST 266	133,200	13,451	888	+2.2	48
CORONET	75,600	13,745	896	+2.1	46

random HCF assignment gives the protection path more low-loss HCF spans, allowing it to outperform the working path despite being physically longer.

Modulation Format Impact (C-Band)

Figure 2 and Table 2 quantify the modulation format impact under Monte Carlo mixed-path deployment.

At 0% HCF (all-SMF, Fig. 2(a)), protection switching between homogeneous SMF paths already causes modulation downgrade due to path-length asymmetry inherent to edge-disjoint routing. At 25% HCF (Fig. 2(b)), 33–67% of node pairs maintain their modulation format, with the remainder experiencing one-step (32–43%) or two-or-more-step (1–21%) downgrades. NSFNET exhibits the highest downgrade rates due to its long inter-node distances.

At 50% HCF (Fig. 2(c)), the picture is broadly similar: 40–60% maintain format. At 75% HCF (Fig. 2(d)), downgrades decrease slightly as more links benefit from HCF's superior GSNR. At 100% HCF (all-HCF, Fig. 2(e)), the fiber-type mismatch between paths vanishes and both paths benefit from HCF's lower loss and nonlinearity, substantially reducing downgrade. In dense, short-path topologies such as Nobel-Germany and COST 239, the GSNR improvement from all-HCF is large enough to push most node pairs well above the modulation threshold, yielding a pronounced drop in downgrade (e.g., Nobel-Germany falls from 37% at 50% HCF to 4% at 100%). Continental-scale topologies (NSFNET, USNET) also show reduced downgrade at 100% HCF, but path-length asymmetry continues to limit some pairs.

Impact of HCF Deployment Fraction

Figure 3 sweeps the HCF deployment fraction from 0% to 100% in 10% steps using the Monte Carlo methodology (200 trials per fraction). Figure 3(a) shows that the mean CD step decreases monotonically with HCF fraction for most topologies. At 0% (all-SMF), CD steps are substantial—NSFNET reaches $\sim 25,000$ ps/nm—

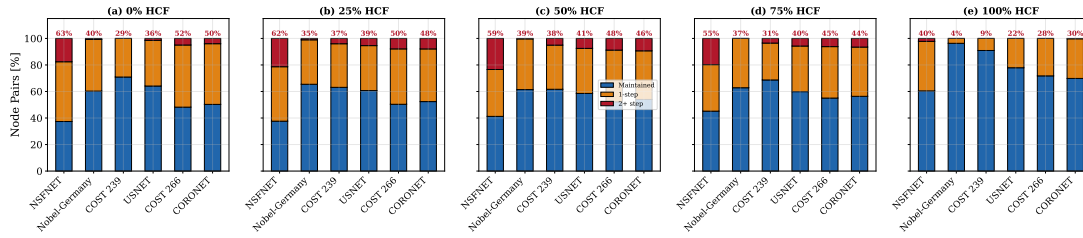


Fig. 2: C-band modulation format impact under Monte Carlo mixed-path deployment at (a) 0%, (b) 25%, (c) 50%, (d) 75%, and (e) 100% HCF fraction (200 trials each). Blue: format maintained; orange: one-step downgrade; red: two-or-more step downgrade. Total downgrade percentage annotated.

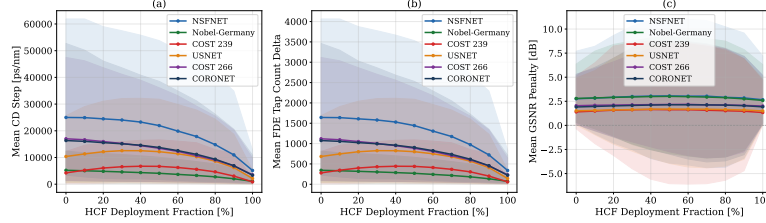


Fig. 3: Monte Carlo HCF deployment sweep (200 trials per fraction): (a) mean CD step generally decreases as HCF replaces high-dispersion SMF (COST 239 rises initially due to inter-path dispersion contrast in its dense topology); (b) mean FDE tap count delta follows the same trend; (c) mean GSNR penalty peaks at intermediate HCF fractions where fiber-type mismatch variance is largest. Shaded bands: 5th–95th percentile.

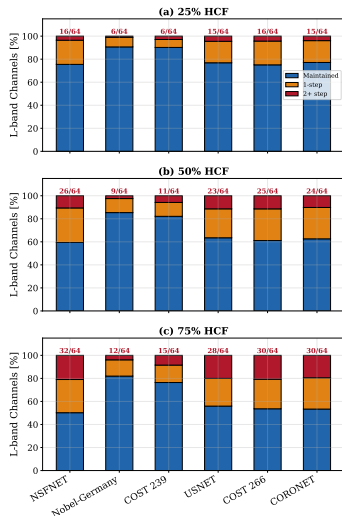


Fig. 4: Fraction of L-band channels affected by CO_2 -induced modulation downgrade under Monte Carlo mixed-path deployment (200 trials each) at (a) 25%, (b) 50%, and (c) 75% HCF. Annotations show mean number of affected channels out of 64.

because $D_{\text{SMF}} = 17 \text{ ps}/(\text{nm km})$ amplifies the inherent path-length differences between working and protection routes. These differences arise from the edge-disjoint constraint in Suurballe’s algorithm: for node pairs connected by a short direct link, the protection path must route the long way around the network, producing path-length differences of several thousand km in continental-scale topologies (e.g., Seattle–Portland in USNET: 280 km working, 3,300 km protection). The wide 5th–95th percentile bands reflect these topology-dependent outliers. COST 239 is a notable exception: its CD step increases from 0% to $\sim 40\%$ HCF before decreasing. This occurs because COST 239 has the densest connectivity (average node degree 4.7, protection/working path-length ratio 1.45) and relatively short paths. At 0% HCF, the small path-length asymmetry produces modest CD steps; introducing a mix of SMF and HCF links creates dispersion contrast within each path, temporarily increasing ΔCD until HCF dominance at higher fractions lowers overall dispersion. At 100% (all-HCF), CD steps are approximately $5\times$ smaller. Figure 3(b) confirms that FDE tap count delta mirrors this trend, since taps scale linearly with CD.

Figure 3(c) shows that the mean GSNR penalty is positive (1.3–3.1 dB) across all fractions, confirming that the working path—always the shorter of the two disjoint

paths—on average achieves higher GSNR. For most topologies, the penalty peaks near 40–50% HCF: at intermediate fractions the random fiber-type mix maximises the variance in per-path GSNR, amplifying the penalty for node pairs where the protection path draws disproportionately more SMF spans. At the extremes (0% and 100%), both paths share the same fiber type and the penalty is governed solely by path-length asymmetry, which is smaller.

L-Band Impairment Analysis

Figure 4 extends the analysis to L-band operation using the independent L-band physical-layer model described in link model. The total L-band penalty relative to C-band comprises two components: (i) a baseline penalty from higher fiber attenuation, EDFA noise figure, and IMI (Table 1), and (ii) the wavelength-dependent CO_2 gas absorption penalty on HCF links.

Figure 4 shows the fraction of L-band channels experiencing CO_2 -induced modulation downgrade under Monte Carlo mixed-path deployment at 25%, 50%, and 75% HCF. For each trial, the per-channel L-band GSNR is computed independently using L-band parameters, and then the per-channel CO_2 penalty (scaling linearly with HCF path length) is subtracted; a channel is considered “affected” if its resulting modulation format drops below what the L-band baseline GSNR would support.

At 25% HCF (Fig. 4(a)), the CO_2 impact is modest: Nobel-Germany and COST 239 show ~ 6 affected channels out of 64, while continental-scale topologies reach 15–16 affected channels. At 50% HCF (Fig. 4(b)), Nobel-Germany shows ~ 9 affected channels, while NSFNET and COST 266 reach 25–26 affected channels due to longer HCF accumulation. At 75% HCF (Fig. 4(c)), the effect intensifies further—NSFNET reaches ~ 32 affected channels—but even for the largest topologies, $\sim 50\%$ of L-band channels remain unimpaired.

Conclusion

Cross-fiber protection switching in hybrid HCF–SMF networks produces CD steps of 4,000–22,000 ps/nm, GSNR penalties of 1.6–3.1 dB, and 38–59% modulation downgrade at 50% HCF deployment. In the L-band, CO_2 absorption downgrades ~ 9 –26 of 64 channels at 50% HCF. While advancing towards an all-HCF architecture generally resolves these switching impairments (except gas absorption), near-term deployments will require fiber-aware routing algorithms and rapid DSP adaptation to manage these extreme contrasts.

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